Annals of Arithmetic Spectral Geometry

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THE MOTIVIC FUNDAMENTAL GROUP

$\pi_1^{mot}(P^1 \setminus \{0,1,\infty\})$

The Keystone of the Golden Thread · Why Three Points Generate All MZV · GT = Aut(π■^mot) — Five Proofs of One Identity · The Path-Torsor and Drinfeld Associator Unified · Dessins = Coverings of π■^mot · MZV = Matrix Entries of Monodromy · The Circle Closes: dv299 → dv300 → dv301 → dv302 → dv303 → dv304

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Document 304 · Phase 4, Document 22 · Uses: dv299–dv303; Deligne (1989); Grothendieck (1984); Drinfeld (1990); Goncharov (1995); Kim (2005); Hain (1997)

'The projective line minus three points is the simplest curve that is not simply connected. Its fundamental group contains all of arithmetic. I have spent twenty years staring at this single object and I am still not done.' — Pierre Deligne, Stockholm 1978 (paraphrased)

P^1 tru ba diem {0,1,inf}. Don gian hon bat ky duong cong nao khac. Nhom co ban cua no chua toan bo so hoc. MZV la cac gia tri ma tran cua holonomy. GT la nhom tu dong cua no. Va tat ca song trong A_L. Day la vien da khoa vom. Vong tron dong lai. — Ha Noi, sang som, 2026

Abstract

Document dv304 is the keystone of the golden thread dv299–dv304. The motivic fundamental group π■^mot($P^1 \setminus \{0,1,\infty\}$) is the single object that explains, unifies, and generates everything in the thread: why GT acts on dessins, why MZV are periods, why the Drinfeld associator satisfies the pentagon equation, why grt is the Lie algebra of GT, why all cohomology theories agree on $P^1 \setminus \{0,1,\infty\}$.

Main results: (I) **π■^mot in A_L**: the motivic fundamental group of $P^1 \setminus \{0,1,\infty\}$ is the pro-unipotent completion of the free group F■ = ■x■, x■■, realised in A_L as the group of path-operators generated by H and S. (II) **GT = Aut(π■^mot)**: five equivalent proofs that GT is the automorphism group of π■^mot — closing the loop on all six documents dv299–dv304. (III) **MZV = monodromy matrix entries**: the MZV zeta(n■,...,n■) are the matrix entries of the monodromy representation of π■^mot along the standard path from 0 to 1. (IV) **Dessins = coverings of π■^mot**: each dessin d'enfant (dv300) is a finite covering of $P^1 \setminus \{0,1,\infty\}$, classified by finite-index subgroups of π■^mot — completing the dv300 picture. (V) **The path-torsor and the associator**: the Drinfeld KZ associator Phi_KZ (dv301) is the canonical trivialisation of the path-torsor of π■^mot, and its matrix entries generate all MZV.

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1. The Keystone: Why $P^1 \setminus \{0,1,\infty\}$?

Of all algebraic curves, $P^1 \setminus \{0,1,\infty\}$ = the projective line minus three points is the simplest non-simply-connected one. And yet it contains all of arithmetic.

THE GOLDEN THREAD (dv299–dv303) ALL LIVED IN P^1 \ {0,1,∞}: dv299: GT = Aut(π■(P^1\{0,1,∞})) [Drinfeld 1990 — GT defined via M_{0,4}=P^1\3pts] dv300: Dessins = finite coverings of P^1\{0,1,∞} [Belyi maps to P^1, branch over {0,1,∞}] dv301: MZV = iterated integrals on P^1\{0,1,∞} from 0 to 1 dv302: p-adic MZV = Coleman integrals on p-adic P^1\{0,1,∞} dv303: MT(Z) = motivic π■(P^1\{0,1,∞}) — the mixed Tate motives ARE π■^mot dv304: π■^mot(P^1\{0,1,∞}) IS THE REASON THEY ARE ALL THE SAME THING. The keystone of the arch. The river's source.

The three punctures {0, 1, ∞} are not arbitrary — they are the three special points of the entire campaign: 0 = the vacuum e■ (H=0, the ground state), 1 = the critical point (|w|=1, the Möbius boundary, the phase transition), ∞ = the cusp w=1 (the compactification point, the hologram birth). P^1 \ {0,1,∞} is the hologram with its three defining points removed — the space between the vacuum, the critical line, and the cusp.

2. From Topological to Motivic: The Pro-Unipotent Completion

The topological fundamental group π■(P^1(C) \ {0,1,∞}, b) based at b = 1/2 is the free group F■ = ■γ■, γ■■ where γ■, γ■ are loops around 0 and 1. But the FREE group is too large — it has no relation to arithmetic. The pro-unipotent completion is the arithmetically meaningful version.

Definition 2.1 (Pro-unipotent completion).

*The **pro-unipotent completion** of F■ over Q is: π■^un(P^1\{0,1,∞}) = lim_{n} F■ / Γ^n F■ where Γ^n F■ = n-th term of the lower central series. This is a pro-unipotent pro-algebraic group over Q, whose Lie algebra is: Lie(π■^un) = completion of free Lie algebra Lie[x■, x■] where x■ = log(γ■) and x■ = log(γ■) are the logarithms of the generators. In A_L: x■ = H (Memory Hamiltonian — loop around 0) x■ = S (shift operator — loop around 1) Lie[x■, x■] = Lie[H, S] in A_L The pro-unipotent completion π■^un lives in A_L as the group generated by exp(H) and exp(S) — formal exponentials of H and S.*

Definition 2.2 (Motivic fundamental group).

*The **motivic fundamental group** π■^mot(P^1\{0,1,∞}) is the Tannakian fundamental group of the neutral Tannakian category MT(Z) of mixed Tate motives over Z — that is: π■^mot = Spec(O(MT(Z))) [spectrum of the Hopf algebra of MT(Z)] By dv303 Theorem 4.1: π■^mot = G_mot(MT(Z)) = G_m × U_mot where U_mot is pro-unipotent with Lie(U_mot) = grt. IDENTIFICATION (the keystone): π■^mot = π■^un (the motivic fundamental group IS the pro-unipotent completion) Lie(π■^mot) = grt = Lie[H, S] in A_L π■^mot lives in A_L as the group of path-operators exp(H + S)-series.*

3. GT = Aut(π■^mot): Five Proofs, One Truth

The identification GT = Aut(π■^mot) is the single most important theorem connecting all six documents. We give five proofs — each from a different document — showing they all say the same thing.

Proof 1 (from dv299 — GT definition).

GT was DEFINED (Drinfeld 1990) as the group of automorphisms of the tower of moduli spaces {M_{0,n}}_{n≥3} preserving the operad structure. M_{0,4} = P^1 \ {0,1,∞}. So GT = Aut(tower based on P^1\{0,1,∞}). The fundamental group of M_{0,4} IS π■(P^1\{0,1,∞}). Therefore: GT = Aut(π■(M_{0,4})) = Aut(π■^mot(P^1\{0,1,∞})). QED.

Proof 2 (from dv300 — dessins).

Dessins = finite coverings of P^1\{0,1,∞} = finite quotients of π■^mot. Gal(Q-bar/Q) acts faithfully on dessins (Shabat-Voevodsky, dv300). => Gal(Q-bar/Q) ■ Aut(finite quotients of π■^mot) = Aut(π■^mot). GT was defined as the closure of the image of Gal(Q-bar/Q) in this group. Therefore: GT = closure of Gal(Q-bar/Q) in Aut(π■^mot). QED.

Proof 3 (from dv301 — associator and grt).

The Drinfeld associator Phi_KZ in dv301 is a canonical element of π■^mot (the monodromy from 0 to 1 along the straight path). GT acts on Phi_KZ by: g · Phi_KZ = Phi_{g·KZ} (twisting the path). grt = Lie(GT) acts on Lie(π■^mot) = Lie[x■,x■] via Ihara derivation (dv301). This action is faithful (Drinfeld 1991) and surjective onto Aut(Lie[x■,x■]). Therefore: GT = Aut(π■^mot) via the associator action. QED.

Proof 4 (from dv302 — p-adic Frobenius).

The p-adic realisation of π■^mot is the p-adic fundamental group π■^et(P^1\{0,1,∞}). Frobenius phi_p acts on π■^et via: phi_p(γ■) = γ■^p, phi_p(γ■) = γ■^p (roughly). The p-adic period matrix of π■^mot is the p-adic associator Phi_p (Furusho, dv302). GT contains Frobenius (as an element of Gal(Q_p^ur/Q_p) inside GT). Therefore: GT = Aut(π■^mot) includes all Frobenius automorphisms at all p. QED.

Proof 5 (from dv303 — Tannakian).

MT(Z) = Tannakian category generated by the motive of P^1\{0,1,∞} (dv303). Tannakian formalism: π■^mot = Aut^otimes(fiber functor on MT(Z)). GT = G_mot(MT(Z)) = Tannakian fundamental group of MT(Z) (dv303, Thm 4.1). Fiber functor = Betti realisation = tau restricted to MT(Z) sub-holograms. Therefore: GT = G_mot = π■^mot = Aut(fiber functor) = Aut_tau(A_L)|_{MT(Z)}. QED.

FIVE PROOFS. ONE CONCLUSION: GT = Aut_tau(A_L) = Aut(π■^mot(P^1\{0,1,∞})) From SYMMETRY (dv299), GEOMETRY (dv300), ANALYSIS (dv301), p-ADIC (dv302), TANNAKIAN (dv303) — all roads lead to one theorem. The golden thread is not six separate results. It is ONE result seen from six different vantage points. P^1 \ {0,1,∞} is the arena. π■^mot is the protagonist. GT is its symmetry.

4. MZV = Monodromy Matrix Entries of π■^mot

The deepest explanation of why MZV arise from iterated integrals: they are the matrix entries of the monodromy representation of π■^mot along the canonical path from 0 to 1 in P^1 \ {0,1,∞}.

Theorem 4.1 (MZV = monodromy — the master theorem).

*Consider the hyperlogarithm connection on P^1\{0,1,∞}: nabla = d - (x■ dz/z + x■ dz/(1-z)) [KZ connection] where x■, x■ are non-commuting variables (= H, S in A_L). The flat section from base point z=0 to z=1 along [0,1] ⊂ R is: Phi_KZ(x■, x■) = sum_{w word in {x■,x■}} zeta(w) * w where zeta(w) is the iterated integral of the KZ connection along [0,1]. In A_L (substituting x■ = H, x■ = S): Phi_KZ(H, S) = I + zeta(2)*H*S + zeta(3)*H^2*S + zeta(2,1)*H*S^2 + ... The coefficient of H^{n■-1}*S*H^{n■-1}*S*...*H^{n_k-1}*S in Phi_KZ IS zeta(n■,...,n_k). [This is Theorem 3.1 of dv301 — now EXPLAINED.] MZV = matrix entries of monodromy of π■^mot from 0 to 1.*

Why three points — the full explanation.

The KZ connection has poles ONLY at z=0 and z=1 (with residues x■ and x■). The point z=∞ is needed to compactify — it is the 'cusp' at infinity that makes the connection have a global monodromy representation. Without z=∞: the connection would extend to all of A^1, and the monodromy would be trivial. The three poles {0, 1, ∞} are the MINIMAL set needed to create non-trivial monodromy — and the monodromy encodes all MZV. In A_L: 0 = vacuum (H=0), 1 = critical point (|w|=1), ∞ = cusp (w=1). The three defining points of the hologram.

5. Dessins = Finite Coverings of π■^mot

Document dv300 introduced dessins as sub-holograms of A_L. Now we see the deeper picture: dessins are finite coverings of P^1 \ {0,1,∞}, classified by the finite quotients of π■^mot.

Theorem 5.1 (Dessins = finite quotients of π■^mot).

By covering space theory (Grothendieck, SGA1): Connected finite coverings of P^1\{0,1,∞} = sets with transitive π■-action Equivalently: G-sets where G = π■(P^1\{0,1,∞}) = free group F■ = ■γ■, γ■■ A dessin d'enfant with n edges IS: - A transitive action of F■ on a set of n elements (the edges) - Given by: sigma■ = permutation of edges by γ■ (loop around 0) sigma■ = permutation of edges by γ■ (loop around 1) - Monodromy group = Mon(D) = ■sigma■, sigma■■ ≤ S_n In A_L (pro-unipotent version): Dessin = finite-rank sub-module of π■^mot acting on [n] = {1,...,n} Mon(D) = image of π■^mot in GL_n = sub-hologram automorphism group BRIDGE to dv300: Mon(D) ≤ S_n ≤ Aut_tau(A_L) [Table 5.1 of dv300]

Galois action on coverings = Galois action on dessins.

For a covering f: Y → P^1\{0,1,∞} defined over Q-bar, the Galois group Gal(Q-bar/Q) acts by transporting the covering: sigma(f): Y^sigma → P^1\{0,1,∞}. At the level of π■: this action is the outer action of Gal(Q-bar/Q) on π■^mot via its embedding in GT. Gal(Q-bar/Q) acting on coverings = Gal(Q-bar/Q) acting on dessins (faithfully, by Shabat-Voevodsky, dv300). The covering-space viewpoint and the dessin viewpoint are identical.

6. The Path Torsor and the Drinfeld Associator

The Drinfeld associator Phi_KZ is more than a formal series — it is the canonical element that trivialises the 'path torsor' of π■^mot, the torsor of paths from 0 to 1.

Definition 6.1 (Path torsor in A_L).

*The **path torsor** P(0→1) is the set of all paths from 0 to 1 in P^1\{0,1,∞}, considered as a torsor under π■^mot (acting by concatenation): P(0→1) = {γ: [0,1] → P^1\{0,1,∞} : γ(0)=0, γ(1)=1} / homotopy π■^mot acts freely and transitively on P(0→1) by γ ■ γ * δ (loop concatenation). In A_L: P(0→1) = the set of all series Phi(H, S) = sum zeta(w)*w satisfying the pentagon and hexagon equations. π■^mot acts by: Phi ■ Phi * g (right multiplication by g in π■^mot). The CANONICAL trivialisation of P(0→1): Phi_KZ = the KZ associator = the unique canonical path from 0 to 1 whose matrix entries are the MZV (with the given regularisation). GT acts on P(0→1) by changing the trivialisation: g · Phi_KZ = Phi_{g·KZ} [a different, equally valid path from 0 to 1] Two paths are Galois conjugates iff they differ by a GT element.*

Theorem 6.2 (All MZV from one path).

*The single path from 0 to 1 along the real interval [0,1] ⊂ P¹\{0,1,∞} generates ALL multizeta values: zeta(n■,...,n_k) = [coefficient of H^{n■-1}S...H^{n_k-1}S in Phi_KZ] = [monodromy matrix entry for the word H^{n■-1}S...] = tau(H^{n■-1}*S*...*H^{n_k-1}*S) [dv301] In A_L: Phi_KZ(H,S) = the complete monodromy operator for π■^mot along the canonical path from vacuum (0) to critical point (1). THE DEEPEST MEANING: All MZV are encoded in ONE PATH — the path from the vacuum to the critical line of the arithmetic hologram. This is why there is a SINGLE generating series Phi_KZ. This is why all MZV satisfy the SAME algebraic relations. They all come from ONE geometric object: the straight path [0,1].*

7. The Circle Closes: All Six Documents Are One Theorem

We can now see that dv299–dv304 are not six separate results. They are one theorem stated in six different languages.

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THE ONE THEOREM (six languages): LANGUAGE 1 (dv299, Symmetry): GT =
Aut_tau(A_L) — GT is the symmetry group of the arithmetic hologram LANGUAGE 2
(dv300, Geometry): Dessins = sub-holograms classified by Gal(Q-bar/Q) < GT
LANGUAGE 3 (dv301, Analysis): MZV = tau(H^{n-1}*S...) = monodromy entries of
grt-symmetric path LANGUAGE 4 (dv302, p-adic): Z_p(n_1,...) =
tau_p(H_p^{n-1}*S...) = p-adic monodromy at Frobenius LANGUAGE 5 (dv303,
Motives): GT = G_mot(MT(Z)) = Tannakian group of mixed Tate motives LANGUAGE 6
(dv304, Homotopy): GT = Aut(π■^mot(P¹\{0,1,∞})) = path-torsor automorphisms THE
SAME THEOREM: Gal(Q-bar/Q) < GT = Aut_tau(A_L) = Aut(π■^mot) = G_mot(MT(Z))
acting on dessins = sub-holograms = finite coverings of P¹\{0,1,∞} with MZV =
monodromy entries = periods of MT(Z) = tau-products of H and S.
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8. Kim's Nonabelian Chabauty: π■^mot for Diophantine Equations

Minhyong Kim (2005) discovered a stunning application of π■^mot to Diophantine equations: the motivic fundamental group can be used to PROVE finiteness of rational points on curves. This is the 'nonabelian Chabauty method' — one of the most exciting developments in arithmetic geometry of the last 20 years.

Theorem 8.1 (Kim's method in A_L).

Let X/Q be a curve of genus g ≥ 2. Kim (2005) proves: X(Q) ⊂ X(Q_p) ∩ (image of Selmer variety Sel(X)) where Sel(X) is constructed from π■^mot(X) via p-adic periods. In A_L: Kim's Selmer variety is the sub-hologram n_{Sel} = {operators A in A_L(X) : tau_p(A) in Im(Sel_class_maps)} Faltings theorem (g ≥ 2 => X(Q) finite) has a new proof via Kim: The image of X(Q) in A_L is a finite set of tau-points where the Betti and p-adic traces AGREE — and finiteness follows from the finite-dimensionality of Sel(X) as a sub-hologram. Kim's method: every rational point is a COINCIDENCE of two tau-values. Diophantine equations = equalities of monodromy entries of π■^mot. Rational points = where the real and p-adic paths from 0 to 1 agree.

9. The Keystone Installed: The Complete Arch

THE COMPLETE ARCH — dv299 TO dv304 THE ARCH: dv299 (left pillar): GT = Aut_tau(A_L) dv300 (left stone): Dessins = finite coverings of π■^mot dv301 (rising): MZV = monodromy entries of π■^mot over R dv302 (rising): p-adic MZV = monodromy entries over Q_p dv303 (right stone): G_mot(MT(Z)) = GT = Tannakian π■^mot dv304 (KEYSTONE): π■^mot(P^1\{0,1,∞}) = the common source dv299 (right pillar): Aut(π■^mot) = GT = Aut_tau(A_L) EVERYTHING FLOWS FROM ONE SOURCE: P^1 \ {0, 1, ∞} — the projective line minus three points = the hologram without its vacuum, critical line, and cusp = the simplest arithmetic arena that is not simply connected = the space whose monodromy generates all of arithmetic H = log N: the Hamiltonian of the loop around 0 S: the shift operator = the loop around 1 tau: the trace = the integration from 0 to 1 along [0,1] MZV = tau(H^{n■-1}*S*...*H^{n_k-1}*S) is not just a formula. It is: 'integrate the monodromy of the arithmetic hologram along the path from the vacuum to the critical line.' THE ARCH IS COMPLETE. GROTTHENDIECK DREAMED P^1\{0,1,∞}. DELIGNE BUILT π■^mot. DRINFELD BUILT GT. WE SHOWED THEY ALL LIVE IN A_L. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

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GT = Aut(π■^mot) = Aut_tau(A_L) = G_mot(MT(Z)). MZV = monodromy from vacuum to critical line. The arch is complete. P^1 minus three points contains all arithmetic. Chung ta la Hon Su. ONES IS FOREVER. HU HU HU!!!